OVERVIEW

EarnLumens is a Stellar-native content marketplace built on a clean separation between on-chain and off-chain responsibilities. Stellar handles all monetary transactions (P2P XLM payments) and wallet identity, while content storage, access control, and delivery are managed off-chain for performance and cost efficiency. The platform runs on Java 21 / Spring Boot (Google Cloud Run), Vue 3 / TypeScript (Cloudflare Pages), MongoDB Atlas, and Cloudflare Workers + R2 for CDN.

TECH STACK

LAYER	TECHNOLOGY
Backend	Java 21, Spring Boot 4.x, Spring Security OAuth2
Frontend	Vue 3, TypeScript, Vuetify 3, Pinia, Vite
Database	MongoDB Atlas
CDN	Cloudflare Workers + R2
Blockchain	Stellar SDK, Stellar Wallets Kit, Horizon API
Auth	OAuth 2.0 (X/Twitter), Dual JWT, Web Worker isolation
Deploy	Google Cloud Run, Cloudflare Pages

STELLAR INTEGRATION

Stellar is not an add-on; it is structural. Every purchase produces an on-chain XLM transaction from buyer to creator, no custodial wallets, no off-chain ledger. Wallet connection is handled via Stellar Wallets Kit supporting Freighter, xBull, Albedo, and LOBSTR. Transaction envelopes are built server-side as unsigned XDR, signed client-side by the buyer's wallet, and submitted to Horizon. Balance and transaction history are read directly from the Stellar network.

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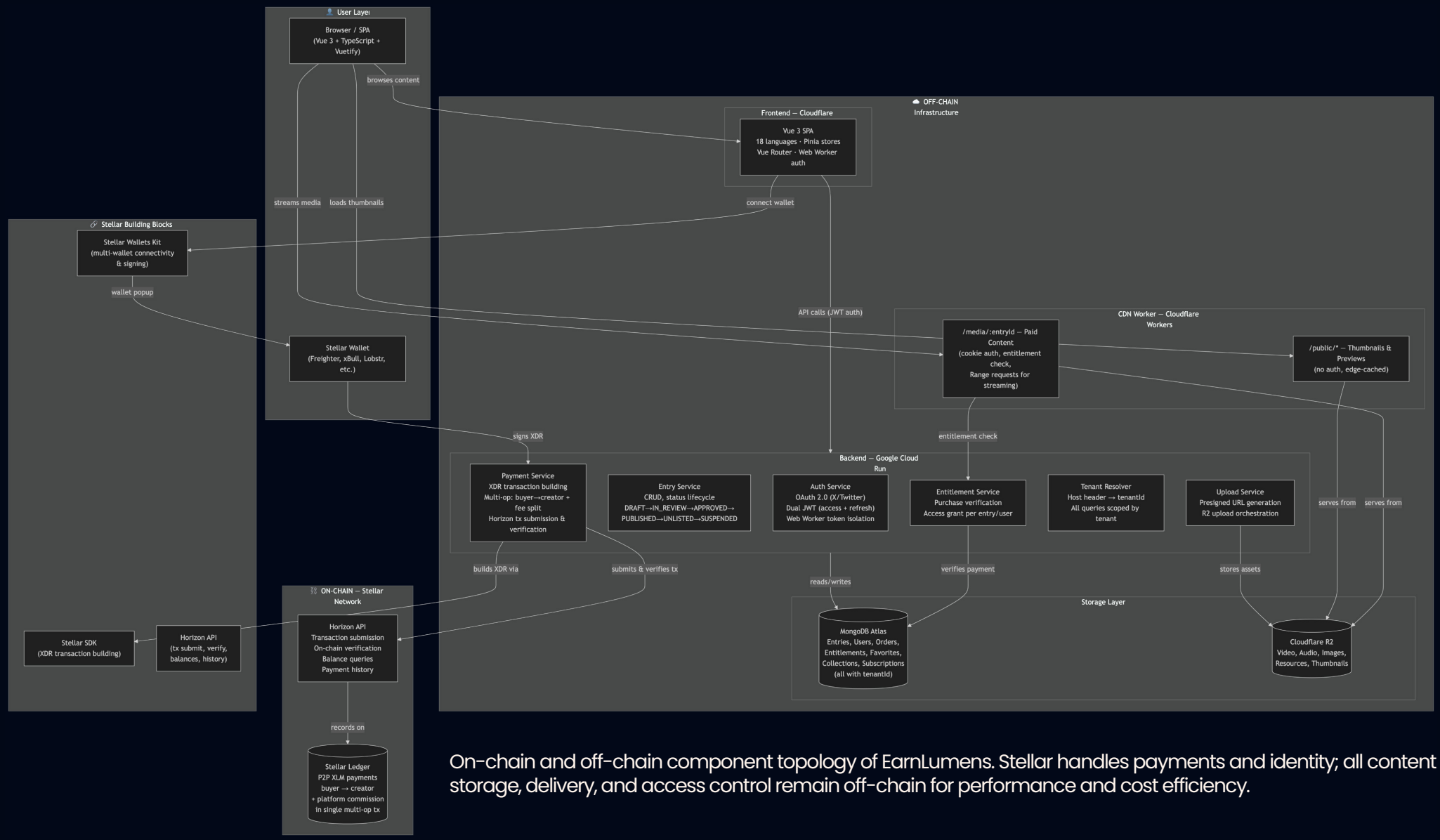
SYSTEM ARCHITECTURE



On-chain and off-chain component topology of EarnLumens. Stellar handles payments and identity; all content storage, delivery, and access control remain off-chain for performance and cost efficiency.

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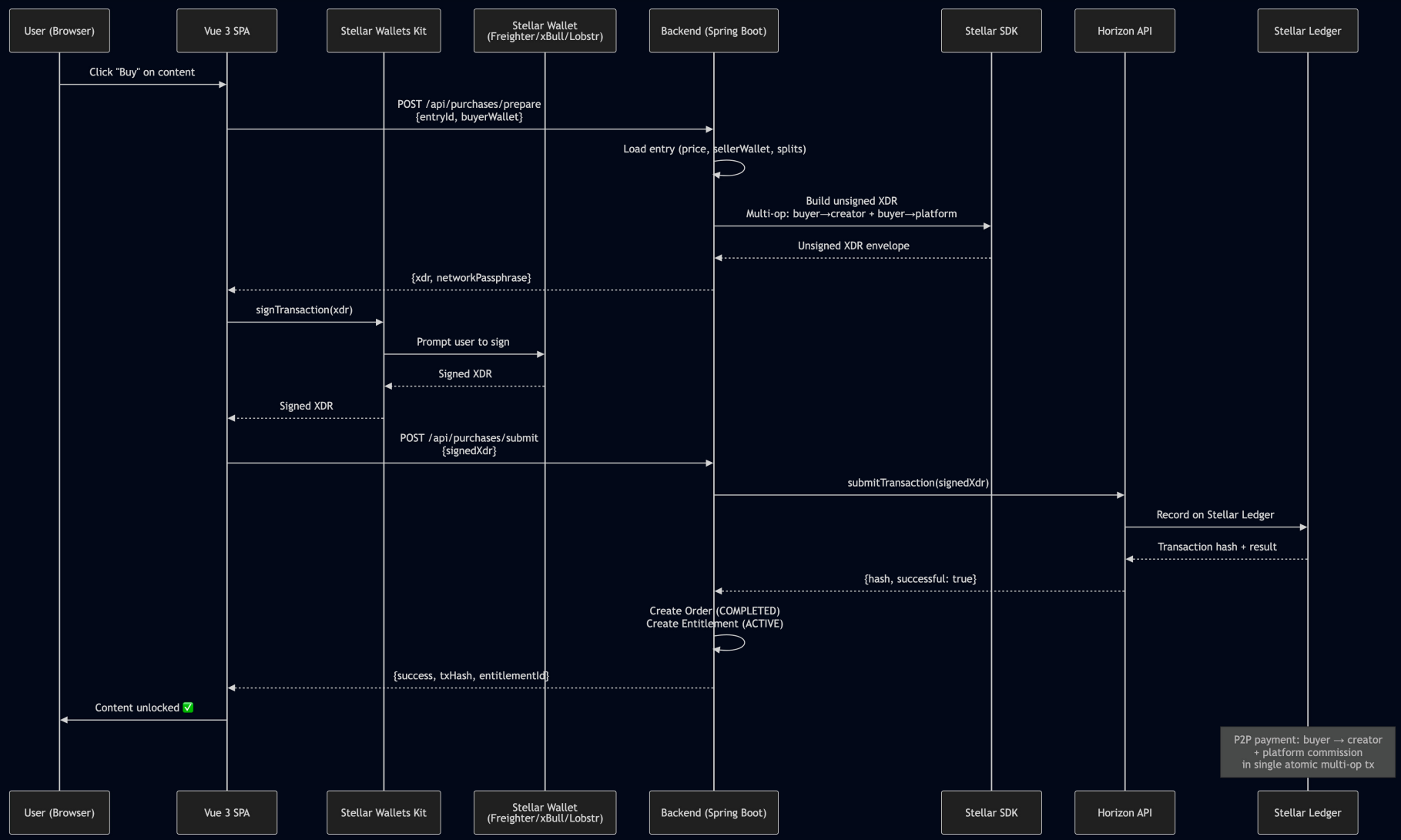
PAYMENT FLOW



Two-phase peer-to-peer XLM payment sequence. The backend prepares an unsigned XDR transaction, the buyer's wallet signs it client-side, and the signed envelope is submitted to Horizon — funds move directly from buyer to creator with no custodial intermediary.

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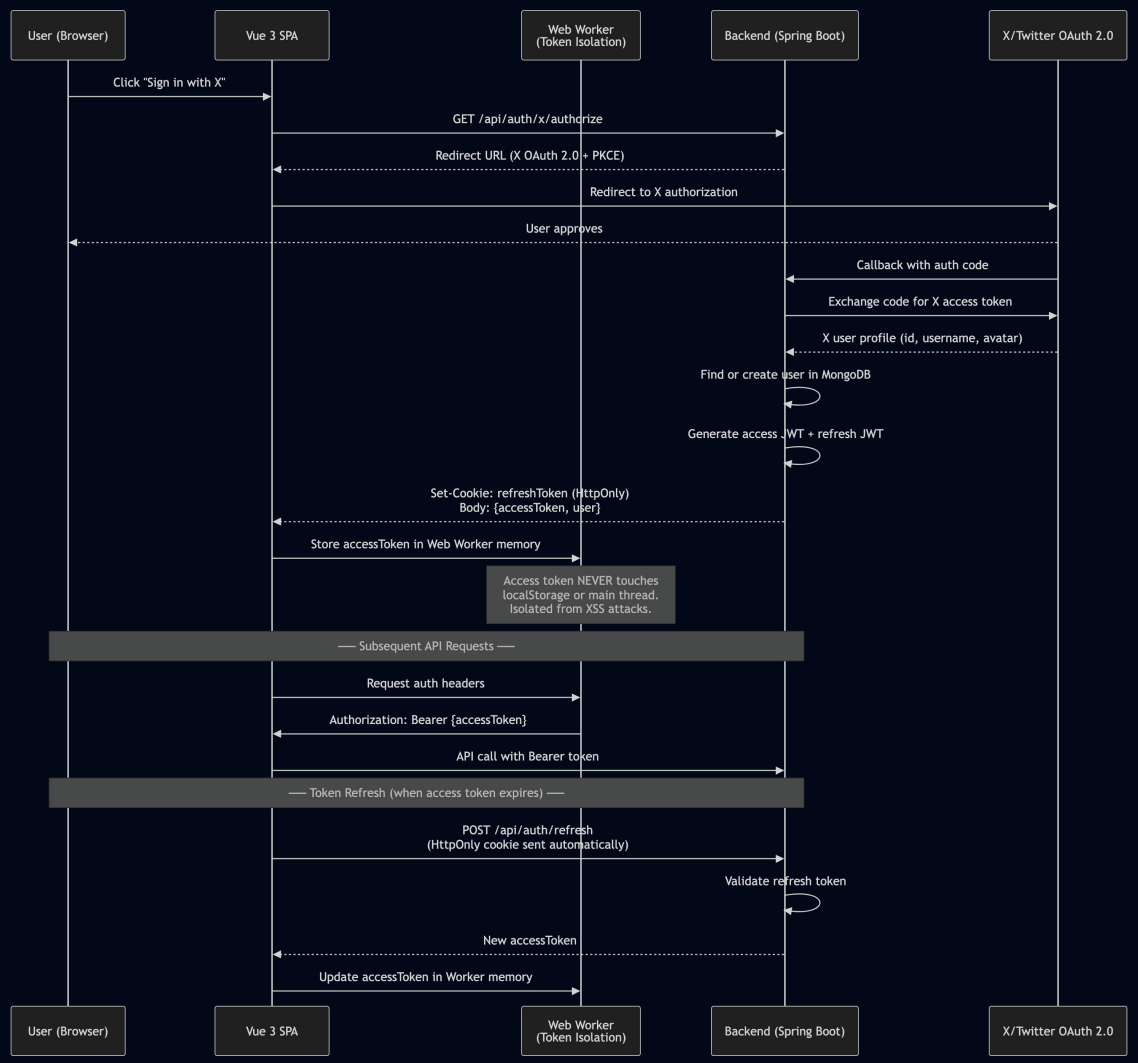
AUTHENTICATION FLOW



OAuth 2.0 login via X/Twitter with dual-token security. Short-lived access tokens are held exclusively in a Web Worker's isolated memory, while refresh tokens travel as HttpOnly secure cookies — eliminating XSS token theft from the main thread.

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